

Proposal for Machine Learning Operations Project

Semester: HS26

Project Title	Swiss Control Area Balance Forecast
Course / Module	MLOPS H26
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Public Repository	GitHub Swiss Control Area Balance Forecast

1 Short Project Description

The project forecasts the imbalance of the Swiss control area (Ausgleichsenergie) using data published daily by Swissgrid, retraining the model every day and serving the latest forecast on a small dashboard.

2 Problem Statement

Problem: A modern problem of electric grids is no longer the production, which has countless options to be sustained, but rather system imbalance. Internal photovoltaic energy production spikes during the day and needs to be sustained at night and during winter. On the external side, neighbouring countries have their own energy overflow, which needs to be contained within an acceptable range to maintain net stability. Energy companies have their internal observations, but normal people do not have a way to predict net stability.

Prediction: Swiss control area imbalance (Total System Imbalance, MW)

Horizon: Next 24h at 15-min resolution, with potentially less accurate prediction for 3-7 days ahead.

Scope: Swiss control area, as published by Swissgrid. In detail, this means one single control area for all of Switzerland, without subdivisions for cantons.

Success criterion: MAE on the 24h-ahead imbalance forecast, computed as $\frac{1}{N} \sum |\hat{y}_t - y_t|$ over 15-min steps.

Baseline: persistence (value from the same quarter-hour one day prior), with its MAE computed directly from historical data before any model training.

Target: beat the persistence baseline of 103.105 MW by a margin of >15 percent.

3 Originality and Motivation

The motivation stems from my work, since we often have to work with energy data, and net stability is a major concern for our customers. Additionally, most of the energy-inspired projects covered on mlops-lab.ch or in the KTH ID2223 primarily focus on PV, which makes this subject doubly suitable.

4 Data sources and features

Source: Swissgrid "Control Area Balance" timeline with daily CSV/XLSX published at 18:30. **History:** Yearly CSVs back to 2009. Used for initial training and to fill out missing values. **Data quality:** Values are only reliable after 6 months, since revisions are possible due to Swissgrid suppliers correcting existing issues. **Label:** Total System Imbalance (MW), independently measured by Swissgrid **Features:** Lagged imbalance (t-1, t-4, t-96, t-672), time-of-day, day-of-week, season, balancing energy price **Optional addition:** MeteoSwiss solar /trepidation data, to test sunlight's effect on imbalance **Leakage:** Out-of-time split: Train date for next day, then evaluate prognosis once complete data arrives. If the retrained model uses the newest data, old prognoses could be improved since the corrected values from newer CSV are included.

5 System Design

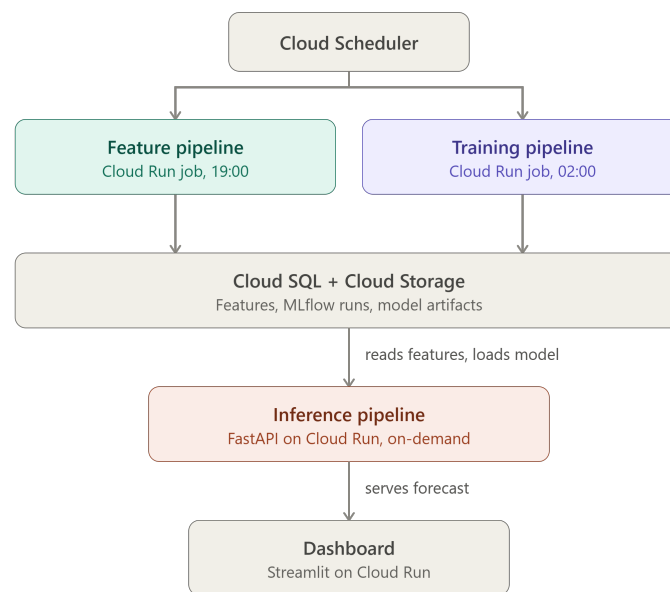


Figure 1: FTI Diagram.

Feature Store: Cloud SQL setup as a single instance with cleaned data as well as MLflow Experiment data. GCS is used to hold MLflow model artefacts, because Cloud SQL isn't optimised for storing entire models.

Experiment Tracking: MLflow tracks daily retraining runs against the baseline. Chosen because the exercises introduced a basic setup.

Orchestration: Google Cloud Scheduler triggers a Cloud Run Job for Python scripts; one for the Feature and Training Pipeline. Chosen since they are serverless triggers, only using credits when called.

Serving: A Cloud Run Service inference pipeline is called from a Streamlit dashboard to serve prognosis data on demand. It will serve as the required public-facing frontend.

Stretch (optional): MeteoSwiss solar/trepidation features to test weather's effect on imbalance. Drift monitoring on the daily retrain.